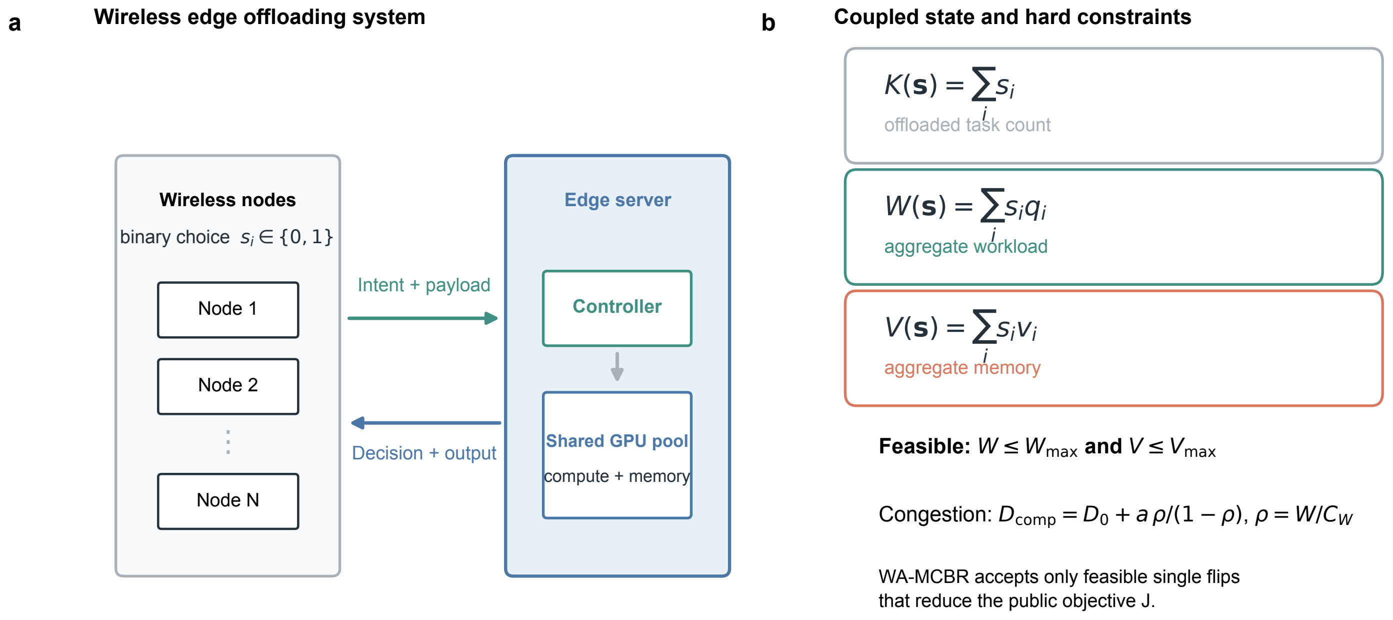
a
Wireless edge offloading system
Wireless nodes
binary choice $s_i \in \{0, 1\}$
Node 1
Node 2
Node N
Intent + payload
Decision + output
Edge server
Controller
Shared GPU pool
compute + memory
b
Coupled state and hard constraints
$K(\mathbf{s}) = \sum_i s_i$
offloaded task count
$W(\mathbf{s}) = \sum_i s_i q_i$
aggregate workload
$V(\mathbf{s}) = \sum_i s_i v_i$
aggregate memory
Feasible: $W \leq W_{\max}$ **and** $V \leq V_{\max}$
Congestion: $D_{\text{comp}} = D_0 + a\,\rho/(1-\rho)$, $\rho = W/C_W$
WA-MCBR accepts only feasible single flips
that reduce the public objective J.